5.13.67

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Question

For what value of k do the following system of equations possess a non trivial (i.e., not all zero) solution over the set of rationals Q?

$$x + ky + 3z = 0 \tag{1}$$

$$3x + ky - 2z = 0 \tag{2}$$

$$2x + 3y - 4z = 0 \tag{3}$$

Theoretical Solution

Given:

$$\begin{pmatrix} 1 \\ k \\ 3 \end{pmatrix}^{\top} \mathbf{x} = 0 \tag{4}$$

$$\begin{pmatrix} 3 \\ k \\ -2 \end{pmatrix}^{\top} \mathbf{x} = 0 \tag{5}$$

$$\begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix}^{\top} \mathbf{x} = 0 \tag{6}$$

Combining (4),(5),(6):

$$A\mathbf{x} = 0 \tag{7}$$

Theoretical Solution

where A is :

$$\begin{pmatrix} 1 & k & 3 \\ 3 & k & -2 \\ 2 & 3 & -4 \end{pmatrix} \tag{8}$$

Theoretical Solution

For the system to have a non-trivial solution, the determinant of the coefficient matrix must be zero. This means the rows are linearly dependent. We simplify the matrix using Gaussian elimination:

$$\begin{pmatrix} 1 & k & 3 \\ 3 & k & -2 \\ 2 & 3 & -4 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - 3R_1} \begin{pmatrix} 1 & k & 3 \\ 0 & -2k & -11 \\ 0 & 3-2k & -10 \end{pmatrix}$$

Now, we continue the row reduction to get the matrix into row echelon form.

Theoretical Solution

Continuing the row reduction:

$$\begin{pmatrix} 1 & k & 3 \\ 0 & -2k & -11 \\ 0 & 3-2k & -10 \end{pmatrix} R_3 \leftarrow 2kR_3 + (3-2k)R_2 \begin{pmatrix} 1 & k & 3 \\ 0 & -2k & -11 \\ 0 & 0 & 2k-33 \end{pmatrix}$$

For a non-trivial solution, the last entry in the third row must be zero.

$$2k - 33 = 0$$
$$\implies k = \frac{33}{2}$$

C Code

```c
double find_k() {
    // A non-trivial solution exists if the determinant of the coefficient
    // matrix is zero. The simplified determinant equation is: -2k + 33 = 0.
    double k_coefficient = -2.0;
    double constant_term = 33.0;
    // We solve for k: k = -constant_term / k_coefficient
    double k = -constant_term / k_coefficient;
    return k;
}
```

Python Code using shared output

```
import numpy as np
import matplotlib.pyplot as plt
import ctypes
import os
# --- Part 1: Call the C Shared Library to
    Find k ---
# Load the shared library
solver_lib = ctypes.CDLL('./5.13.67.so')
# Specify the function's return type to
    ensure it's read as a float
solver_lib.find_k.restype = ctypes.c_double
# Call the C function to get the exact
    value of k
k = solver_lib.find_k()
```

Python Code using shared output

```
# --- Part 2: Plotting the Three Planes
    Using the Result from C ---
# Create a figure and a 3D axes object
fig = plt.figure(figsize=(12, 10))
ax = fig.add_subplot(111, projection='3d')
# Create a grid of points for the surfaces
x_grid = np.linspace(-10, 10, 50)
y_grid = np.linspace(-10, 10, 50)
X, Y = np.meshgrid(x_grid, y_grid)
# Equations of the three planes, using the
    value of k from the C library
Z1 = (-X - k*Y) / 3
Z2 = (3*X + k*Y) / 2
Z3 = (2*X + 3*Y) / 4
```

```python
# Plot the surfaces of the three planes
ax.plot_surface(X, Y, Z1, alpha=0.5, color='royalblue')
ax.plot_surface(X, Y, Z2, alpha=0.5, color='crimson')
ax.plot_surface(X, Y, Z3, alpha=0.5, color='forestgreen')
# Plot the line of intersection (its direction vector is constant)
t = np.linspace(-2, 2, 100)
ax.plot(15 * t, -2 * t, 6 * t, color='black', linewidth=4)
# --- Formatting the plot ---
ax.set_xlabel('X-axis', fontsize=12)
ax.set_ylabel('Y-axis', fontsize=12)
ax.set_zlabel('Z-axis', fontsize=12)
ax.set_title(f'Intersection of Three Planes at k = {k:.1f} (from C library)', fontsize=16, fontweight='bold')
```

Python Code using shared output

```
import matplotlib.patches as mpatches
from matplotlib.lines import Line2D
p1 = mpatches.Patch(color='royalblue', label=f'x + {k:.1f}y + 3z = 0')
p2 = mpatches.Patch(color='crimson', label=f'3x + {k:.1f}y - 2z = 0')
p3 = mpatches.Patch(color='forestgreen', label='2x + 3y - 4z = 0')
line = Line2D([0], [0], color='black', linewidth=3, label='Line of Intersection')
ax.legend(handles=[p1, p2, p3, line], fontsize=10)
plt.show()
```

Python Code

```python
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D
# Define the value of k for a non-trivial solution
k = 33/2
# Create a meshgrid for x and y values
x_vals = np.linspace(-5, 5, 50)
y_vals = np.linspace(-5, 5, 50)
X, Y = np.meshgrid(x_vals, y_vals)
# Calculate Z values for each plane
Z1 = (-X - k * Y) / 3
Z2 = (3 * X + k * Y) / 2
Z3 = (2 * X + 3 * Y) / 4
```

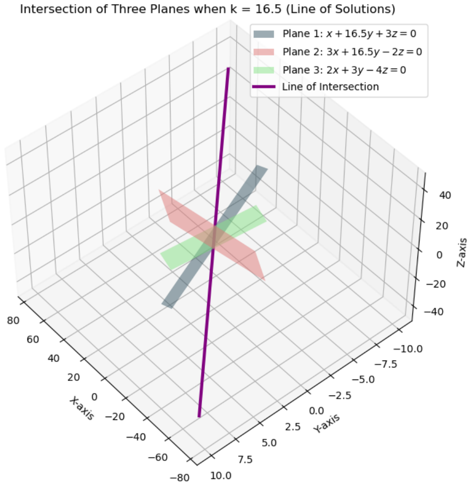
Python Code

```
# Setup the 3D plot
fig = plt.figure(figsize=(10, 8))
ax = fig.add_subplot(111, projection='3d')
# Plot the three planes
ax.plot_surface(X, Y, Z1, alpha=0.5, rstride=100, cstride=100, color='skyblue', label='Plane 1')
ax.plot_surface(X, Y, Z2, alpha=0.5, rstride=100, cstride=100, color='lightcoral', label='Plane 2')
ax.plot_surface(X, Y, Z3, alpha=0.5, rstride=100, cstride=100, color='lightgreen', label='Plane 3')
t = np.linspace(-5, 5, 100) # Parameter for the line
line_x = 15 * t
line_y = -2 * t
line_z = 6 * t
```

Python Code

```
ax.plot(line_x, line_y, line_z, color='
    purple', linewidth=3, label='Line of
    Intersection')
# Set labels and title
ax.set_xlabel('X-axis')
ax.set_ylabel('Y-axis')
ax.set_zlabel('Z-axis')
ax.set_title(f'Intersection of Three Planes
     when k = {k} (Line of Solutions)')
ax.legend(['Plane 1: $x + {:.1f}y + 3z = 0$
    '.format(k),
'Plane 2: $3x + {:.1f}y - 2z = 0$'.format(k
    ),
'Plane 3: $2x + 3y - 4z = 0$',
'Line of Intersection'])
# Adjust view for better visibility
ax.view_init(elev=20, azim=-45)
plt.grid(True)
plt.show()
```

Plot by python using shared output from c



Plot by python